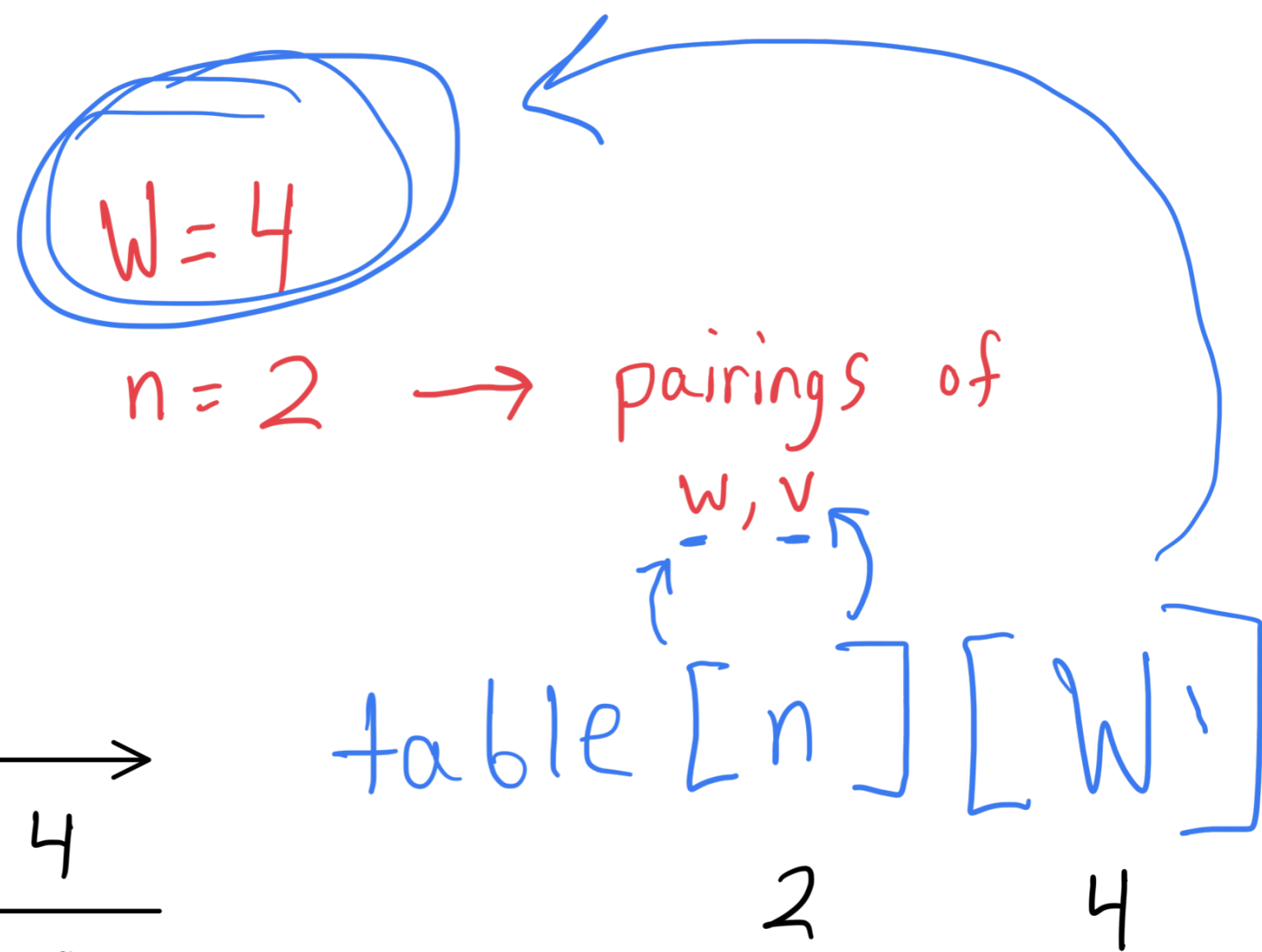


Skyrim

weights = $\{1, 3\}$

v gold = $\{5, 2\}$



		0	1	2	3	4
$n \{w_n, v_n\}$	0	0	0	0	0	0
$n_1 \{1, 5\}$	1	0	5	5	5	5
$n_2 \{3, 2\}$	2	0	5	5	5	7

1. Subprob

W = total weight capacity

$n_{(w,v)}$ = pairing where w_n represents weight of gold and v is price of gold

table $[n][W]$
 | |
 rows columns

2. Recursive Sol?

base case

if ($w_n > W$) return table $[n-1][W]$

recursive case

return max ($v_n +$ table $[n-1][W - w_n]$,
 table $[n-1][W]$) + recurse call (W, n, table)

Runtime?

The runtime would be $O(nW)$ because you're checking each pair of $\{w_n, v_n\}$ against the total weight capacity W to get a subset of n that gives you the value closest to W .